Exp No:5 Data Collection and Initial Exploration
Objective: To collect, load, and perform initial exploration of the diabetes dataset.

PROGRAM:

```
import pandas as pd

import matplotlib.pyplot as plt

# Create sample dataset

data = {

  "ProductID": [201, 202, 203, 204, 205],

  "Quantity": [15, 9, 6, 14, 11],

  "Price": [120.50, 250.75, 175.00, 80.25, 350.00]

}

# Create DataFrame

df = pd.DataFrame(data)

# Display dataset

print("DATASET:")

print(df)

# Dataset information

print("\nDATASET INFORMATION:")

df.info()

# Summary statistics

print("\nSUMMARY STATISTICS:")

print(df.describe())

# Check for duplicates

print("\nDUPLICATES:")

print(df.duplicated().sum())

# Histogram without grid

df.hist(bins=10, figsize=(10, 6))

plt.grid(False)

plt.show()
```

OUTPUT:

```
DATASET:
   ProductID  Quantity   Price
0        201        15  120.50
1        202         9  250.75
2        203         6  175.00
3        204        14   80.25
4        205        11  350.00

DATASET INFORMATION:
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 3 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   ProductID  5 non-null      int64
 1   Quantity   5 non-null      int64
 2   Price      5 non-null      float64
dtypes: float64(1), int64(2)
memory usage: 192.0 bytes

SUMMARY STATISTICS:
        ProductID   Quantity      Price
count    5.000000   5.000000    5.00000
mean   203.000000  11.000000  195.30000
std      1.581139   3.674235  107.52956
min    201.000000   6.000000   80.25000
25%    202.000000   9.000000  120.50000
50%    203.000000  11.000000  175.00000
75%    204.000000  14.000000  250.75000
max    205.000000  15.000000  350.00000

DUPLICATES:
0
```





